

Nicholas Molargik

nick@molargiksoftware.com | molargiksoftware.com | 260-226-9022 | linkedin.com/in/nicholas-molargik | Fort Wayne, IN

EDUCATION

Purdue School of Engineering and Technology, IUPUI - Bachelor's of Science, **Electrical Engineering**, Minor in Mathematics - GPA: 3.72 - Graduated in May 2020.

Codecademy.com - Coursework in HTML, CSS, JavaScript, Java, Python, React Native, Swift, Kotlin, SQL

EXPERIENCE

VT Sports

- Software Engineer | 10/2024 - Present

- Contractor, Lucid Services Group - Mobile Software Engineer | 06/2024 - 10/2024

- Built iOS shot tracer app by **training a custom computer vision / small-object detection model** trained on 28,000 golf ball images and converted for execution on Apple platforms
- Designed **multi-phase tracking algorithm with trajectory-based detection gating and cluster rejection**
- Delivered **production video pipeline processing frames at 120 FPS** with real-time shot trace rendering
- Developed **cross-platform video composition features using Kotlin Multiplatform, Swift, AVFoundation, and FFmpeg** supporting both iOS and Android, resulting in improved performance in playback and stitching.
- Developed **camera capture experiences using Swift, UIKit, Objective-C, AVFoundation, and CoreGraphics**, packaged as a **Flutter Native Plugin** deployed via Codemagic, creating efficiency improvements in core features and CICD workflows.
- **Integrated hardware and software systems** by connecting a new pressure mat device to the mobile experience via a **custom C library**, generating heat map visuals and provided real-time biomechanics feedback to the user.
- **Authored comprehensive technical documentation** covering system architecture, data flow, and method call chains to ensure long-term maintainability, enabling rapid onboarding across proficiencies.
- **Led cross-functional discussions** and drove major software refactors to implement new features, improve code architecture, and improve performance by streamlining integrations, ultimately resulting in faster feature deployment.

Molargik Software LLC

- Founder, Developer | 07/25 - Present | Apps: Mygra · Waffle · Stork · SetDeck (all published on the App Store)

- Architected, designed, and **shipped five complete Swift-based multiplatform apps** across domains including health analytics, web browsing, utilities, and habit tracking.
- Built **advanced SwiftUI interfaces** with custom animations, navigation stacks, liquid glass, and multi-window support for iPad.
- Designed persistent data models using **SwiftData + CloudKit**, enabling real-time device sync.
- Integrated Foundation **Models, HealthKit, and WeatherKit** for on-device intelligence, physiological analytics, and personalized insights.
- Created rich **SwiftCharts** visualizations and offline-first performance using **Swift Concurrency and CloudKit** transactions.
- Delivered deep platform integration—**Widgets, Live Activities, watchOS complications, Siri Intents**, etc.

Sweetwater Sound

- Software Engineer, Mobile & Desktop Applications, 4D Relational Databases | 04/2022 - 04/2024

- **Transformed Sweetwater Sound's mobile applications into world-class storefronts** and utilities by designing brand new codebase architecture and UI. Built two native applications from scratch to replace an abysmal Flutter application. **Significantly improved discovery-to-sales** conversion rates and user satisfaction, ultimately overtaking their website in sales.
- **Mentored senior employees** in macOS application development with **Swift and SwiftUI**.
- Expanded proficiencies in Swift, SwiftUI, Kotlin, Jetpack Compose, SQL, relational databases, unit testing, CICD, Kanban, and lean development techniques, alongside exposure to many other technologies.
- **Deployed iOS and Android applications** to their respective stores to increase rate of value delivery.
- Led design and development of major software projects and user features
- Presented project achievements and future goals to large department groups resulting in broadened understanding and utilization.

STATEMENT OF EXCELLENCE

Innovative and results-driven software engineer specializing in Swift, SwiftUI, and cross-platform development. Founder of Molargik Software, I design and ship advanced native iOS apps and have led major mobile rebuilds—revitalizing legacy codebases into performant, maintainable, and user-focused experiences across Apple and Android platforms.